Shuo Sun

Assistant Professor, Industrial & Systems Engineering
University of Wisconsin–Madison
ssun279@wisc.edu https://shuo-ss.github.io

Research Interests

Modeling and algorithm design for supply chain and revenue management using optimization and machine learning

Academic Appointment

University of Wisconsin–Madison *07/2026 – Current*
Assistant Professor, Department of Industrial & Systems Engineering

Education

University of California, Berkeley *08/2021 – 07/2026*
Ph.D. in Industrial Engineering and Operations Research
Minor in Statistics & Algorithm Design
Advisors: Zuo-Jun Max Shen & Rajan Udwani

Tsinghua University *08/2016 – 06/2021*
B.E. in Industrial Engineering

University of Wisconsin-Madison *08/2019 – 12/2019*
Exchange student in the Department of Industrial and Systems Engineering

Industry Experience

Amazon, Seattle, WA *05/2024 – 08/2024*
Applied scientist intern at Fashion, Fitness & Creators

Honors & Awards

Honorable mention, POMS-HK Best Student Paper Competition *2025*
Winner, INFORMS Daniel H. Wagner Prize *2024*
Finalist, INFORMS RMP Jeff McGill Best Student Paper Award *2024*
IEOR Ph.D. First-year Fellowship *2021–2022*
Finalist, INFORMS Undergraduate Operations Research Prize *2021*

Preprints & Publications

[1] "A Markovian Approach for Cross-Category Complementarity in Choice Modeling", with Omar El Housni and Rajan Udwani. Submitted

- Accepted by the *22nd Conference on Web and Internet Economics (WINE 2026)*

[2] "A Unified Algorithmic Framework for Dynamic Assortment Optimization under MNL Choice", with Rajan Udwani and Zuo-Jun Max Shen. Major revision at *Operations Research*

- *Proceedings of the 26th ACM Conference on Economics and Computation*
- **Finalist, 2024 INFORMS RMP Jeff McGill Student Paper Award**
- **Honorable Mention, 2025 POMS-HK Best Student Paper Competition**

[3] "JD.com Improves Fulfillment Efficiency with Data-driven Integrated Assortment Planning and Inventory Allocation", with Zuo-Jun Max Shen, Yongzhi Qi, Hao Hu, Ningxuan Kang, Jianshen Zhang, Xin Wang, Xiaoming Lin. *INFORMS Journal on Applied Analytics, 2025*

- **Winner of the 2024 INFORMS Daniel H. Wagner Prize**

[4] "Online MDP with Prototypes Information: A Robust Adaptive Approach", with Meng Qi and Zuo-Jun Max Shen. *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence, 2025*

Teaching

Graduate Student Instructor, Economics of Supply Chains Spring 2023, 2025, 2026
Graduate Student Instructor, Supply Chain & Logistics Management Spring 2024
Graduate Student Instructor, Optimization Analytics Fall 2022, 2023

Services

Reviewer for Management Science, Operations Research, Manufacturing and Service Operations Management, Production and Operations Management, MSOM SIG

Session Chair for INFORMS RMP 2024, INFORMS APS 2025, INFORMS Annual Meeting 2025

Conference Talks

"Dynamic Assortment Optimization Under Uniform Prices and Arbitrary Demand Distributions"

- INFORMS RMP 2026

"Resource-Backed Bayesian Persuasion"

- POMS Annual Meeting 2026

"A Markovian Approach for Cross-Category Complementarity in Choice Modeling"

- INFORMS RMP 2025, INFORMS Annual Meeting 2025, POMS Annual Meeting 2026

"A Unified Algorithmic Framework for Dynamic Assortment Optimization under MNL Choice"

- INFORMS Annual Meeting 2023, INFORMS RMP 2024, POMS-HK 2025, INFORMS APS 2025, Purdue Supply Chain and Operations Management Conference 2025

"JD.com Improves Fulfillment Efficiency with Data-driven Integrated Assortment Planning and Inventory Allocation"

- INFORMS Analytics 2025 (award session)

"Online MDP with Prototypes Information: A Robust Adaptive Approach"

- ISMP 2024, AAAI 2025 (poster presentation)

Other Experience

Member, Berkeley IEOR Grad Student Services Leadership Group	*2024, 2025*
Panelist, Berkeley IEOR Ph.D. Visiting Day	*2023, 2024*
Mentor for new Ph.D. students, Berkeley IEOR	*2023*
Panelist, Berkeley IEOR PhD Orientation Day	*2022*